

Validation-Selected Empty-Window Recommendations: An Offline Post-Bin Occupancy-Forecasting Case Study

Anonymous Author(s)

Author information withheld for review

Abstract—Occupancy forecasts are useful for building operations only when their information timing and downstream decision are made explicit. This paper analyzes saved outputs from a single-building, offline post-bin case study at 15-min resolution. The historical schedule baseline, LightGBM, and a compact Transformer encoder produce uncalibrated Empty-class scores; their convex blend is selected on validation AUPRC, and its score threshold is selected separately on validation to maximize offline safe load opportunity subject to an empirical interval conflict cutoff and a one-hour stable-window rule. The stored anchor label t denotes the left label of a completed input bin $[t, t + 15 \text{ min})$, so its effective availability boundary is $t + 15 \text{ min}$. Accordingly, the policy’s midnight-labelled anchors are not claims of recommendations available at 00:00. Across 388,032 overlapping held-out forecast rows, the selected blend has Empty AUPRC 0.8514, versus 0.8497 for the historical baseline and 0.8382 for LightGBM. Across 43 non-overlapping midnight-labelled policy horizons, whose effective boundary is 00:15, its fixed rule recommends 259 intervals; all are subsequently camera-label-empty and form 14 label-safe windows, coinciding with 490.1 kWh of processed HVAC-plus-lighting load proxy. LightGBM coincides with 493.9 kWh but has 11 camera-label conflicts. Validation perturbations show that neither the blend weights nor the threshold is uniquely stable. The result is therefore evidence about reproducible saved-output, post-bin opportunity accounting—not verified energy savings, comfort, physical absence, calibrated risk, real-time availability, or controller performance. Source-side imputation provenance and raw-stream timestamps remain unresolved; a prospective empirical rerun is required for operational claims.

Index Terms—occupancy forecasting, building energy management, offline load opportunity, post-bin evaluation, stable empty windows, validation-selected recommendation

I. Introduction

Building energy-flexibility research considers how operational demand can be adjusted while respecting service constraints [1], [2]. Occupancy is an important contextual signal for that task: it can identify intervals in which a building zone may be empty and, therefore, potentially suitable for a carefully constrained operational action. Occupancy modeling has long been studied for building energy management [3], and recent reviews show a broad forecasting literature spanning data acquisition, estimation, prediction, and evaluation [4]–[6]. Yet a good occupancy forecast alone does not answer whether a recommendation can be made at a useful time, for a useful duration, and at an acceptable observed conflict rate.

That distinction matters because the outcome appropriate to a forecasting study is not automatically an energy outcome. Occupancy-prediction-based cooling-control studies and HVAC-control evaluations can assess a modeled or field operating intervention [7]–[9]. Such evidence is stronger than a retrospective coincidence between a score, a camera-derived Empty label, and a processed meter-derived load proxy. This paper deliberately studies the latter: an offline safe load opportunity. It sums the HVAC-plus-lighting proxy during intervals that a fixed rule recommends and that are subsequently camera-label-empty. It is neither a counterfactual estimate of energy saved nor a claim that the same load can be shifted without comfort, equipment, or control constraints.

The analysis also makes an otherwise easy-to-miss timing distinction explicit. The repository’s 15-min records are left-labelled. A row labelled 00:00 summarizes the input bin $[00:00, 00:15)$ and is therefore treated here as available only at 00:15. This converts the existing evidence into a retrospective, post-bin next-record analysis; it cannot support a claim that a recommendation was available at 00:00. The upstream input is a cleaned release, and neither original raw streams nor source-side imputation provenance are committed. Thus, post-import row ordering is checked, but fully prospective data availability is not established.

The bounded research question is: for one office-building time series, how do validation-selected Empty-class scores trade off forecast discrimination, subsequently observed camera-label conflict, and offline safe load opportunity when recommendations must form at least one-hour stable windows? The evaluation separates two scopes that are often conflated. Forecast discrimination is measured on all overlapping rolling 24-h forecasts. In contrast, the operating rule is selected and evaluated only on non-overlapping midnight-labelled daily horizons, so policy accounting does not double count repeated predictions for the same interval.

The paper makes three contributions. First, it documents an explicit post-bin chronological protocol that maps 24-h Empty-class scores to stable-window recommendations without using controllable loads as predictor inputs. Second, it compares a historical schedule baseline, LightGBM, and a validation-selected score blend using

both AUPRC and a validation-constrained offline policy outcome. Third, it exposes the uncertainty and boundary conditions behind the point estimates: the selected blend has a slightly higher AUPRC than the two baselines and zero observed camera-label conflicts in the held-out policy period, but paired test contrasts cross zero and validation perturbations do not identify unique weights or a unique threshold. The contribution is an evaluative saved-output case study, not a deployed controller, a real-time decision system, or a decision-focused learning algorithm.

Section II positions this evaluation against occupancy forecasting and control research. Sections III and IV define the data, timing, policy, and experimental protocol. Section V reports canonical results, and Sections VI–VII discuss their limits and implications. Appendix A records the status of the expanded decision-aware and window-aware searches without using them to support the primary conclusion.

II. Related Work

A. Occupancy Forecasting

Occupancy forecasting is a mature building-analytics problem rather than a new modeling task. Reviews by Jin et al. and Caballero-Peña et al. organize the literature around forecasting applications, data sources, and evaluation processes [4], [5]. Earlier commercial-building comparisons include stochastic and machine-learning approaches [10], while work based on indoor environmental measurements illustrates the potential and data-quality dependence of sensor-assisted prediction [11]. Sequence models also remain active in the field, including Transformer-based occupancy prediction [12]. These studies motivate a broad baseline set, but they do not establish that one model family is uniformly preferable for every operating rule or building.

The present study accordingly retains a train-only schedule prior alongside nonlinear tabular and sequence models. The tabular reference uses LightGBM [13], the random-forest comparator follows Breiman [14], and the sequence comparator is a compact encoder-only Transformer with a known-future calendar projection, architecturally motivated by [15]. The saved artifact identifier DLinear is a direct one-layer occupancy-history baseline, not the decomposition-based DLinear architecture of Zeng et al. [16]; it is retained only as a simple linear comparator and is relabelled accordingly in the paper. The goal is not to claim algorithmic novelty for any component. Instead, the comparison tests whether combining distinct saved score forecasts produces an inspectable, validation-selected offline policy under the same target and time split.

B. From Occupancy Prediction to Building Operation

Several studies couple occupancy predictions to a downstream HVAC or ventilation decision. Peng et al. evaluate occupancy-prediction-based cooling control [7]. Esrafilian-Najafabadi and Haghighat analyze how occupancy-model

choices relate to HVAC control performance [8], and Zhuang et al. develop probabilistic occupancy forecasts for risk-aware ventilation [17]. Wietzke et al. further connect occupancy prediction to model predictive control and simulation [18]. Those papers demonstrate why a forecast-only score may not summarize a downstream trade-off. They also delimit this paper’s evidence: it does not execute an HVAC action, run a thermal simulation, or measure a counterfactual energy or comfort outcome.

The distinction is especially important when comparing offline studies with operational-system evaluations. Lin et al. report an occupancy-based HVAC control-system evaluation in an office building [9]. This paper has no installation, occupant study, or measured before-and-after intervention. It therefore uses the term opportunity only for a processed meter-derived load proxy coinciding with a recommended and camera-label-empty interval, and it reserves claims about savings or control performance for future intervention or simulation work.

C. Decision-Layer Evaluation

Decision-focused learning and predict-then-optimize methods train predictive models against a downstream decision objective [19], [20]. The present pipeline does not do this: model parameters are not trained with a differentiable decision loss. It fixes candidate score models, selects blend weights by validation AUPRC, and selects a score threshold by an empirical validation conflict cutoff and offline opportunity objective. The appropriate description is therefore a validation-selected decision-layer evaluation, not a calibrated risk guarantee or an implementation of decision-focused learning.

III. Methodology

A. Data, Target, Timing, and Information Set

The source is the cleaned LBNL Building 59 release described by Luo et al. [21] and distributed through Dryad [22]. The repository retains one selected south-zone subset and resamples the imported streams to 15-min bins. An interval receives the camera-derived occupied label when the maximum selected south-zone camera count is greater than zero. The analysis flips this label so that $y_{d,t} = 1$ denotes Empty and $y_{d,t} = 0$ denotes camera-labeled occupied. Thus, Empty means no detected occupant in the selected camera counts, not verified physical absence. The analysis contains 26,413 local-time rows from May 2018 through February 2019. Table I gives the exact split and policy scopes.

Naive source timestamps are interpreted as UTC and converted to America/Los_Angeles before time features are formed. A solar-radiation peak sanity check supports this interpretation, but the source documentation does not explicitly state the timestamp timezone. More importantly, each resampled record uses an explicit left-closed, left-labelled convention: its label t denotes $[t, t + 15 \text{ min})$. The tabular and sequence models consume the anchor

TABLE I
Data and canonical evaluation protocol.

Item	Canonical setting
Data scope	Cleaned Building 59 selected south-zone streams; 26,413 15-min local-time rows, 22 May 2018–21 Feb. 2019 02:00
Empty target	Camera-derived Empty = 1 iff the maximum selected south-zone count is zero in the 15-min bin
Chronological splits	Train: 23 May–25 Nov. 2018 (32.65% Empty); validation: 28 Nov. 2018–6 Jan. 2019 (29.64%); test: 9 Jan.–21 Feb. 2019 (27.75%)
Boundary separation	24.25-h train-validation and validation-test gaps
Input-bin timing	Anchor label t denotes completed left-labelled $[t, t + 15 \text{ min})$ bin; effective boundary $t + 15 \text{ min}$
Forecast scope	96 history and 96 target steps; targets $[t + 15 \text{ min}, t + 24 \text{ h} + 15 \text{ min})$; all overlapping test forecasts
Policy scope	39 validation and 43 test midnight-labelled, non-overlapping horizons; effective test boundary 00:15
Stable recommendation	Contiguous Empty score at or above threshold for ≥ 4 steps (1 h), extracted independently in each policy horizon
Opportunity proxy	Post-import row-order-filled max(HVAC _S + lighting _S , 0) $\times$ 0.25 h only in recommended-and-camera-label-empty intervals
Source limit	Original stream timestamps and upstream imputation lineage unavailable; no prospective availability claim

record, so this paper treats its effective availability boundary as $t + 15 \text{ min}$, after the completed bin. Its first target bin begins at that boundary; a 96-step horizon spans $[t + 15 \text{ min}, t + 24 \text{ h} + 15 \text{ min})$. A policy anchor labelled 00:00 therefore has an effective boundary at 00:15 and targets the following 96 bins, not a recommendation available at 00:00.

Calendar variables are known for future target times. The tabular information set comprises past occupancy/count lags and rolling summaries, completed-input-bin Wi-Fi, temperature, and weather values, and target-time calendar variables. After import, environmental and association-count sensors are forward-filled in row order only; leading missing values are set to zero. This operation prevents new backward fills within the repository, but it cannot establish causal provenance for imputation already present in the cleaned release. Wi-Fi has 81.57% missingness before this fill and no staleness cap or missingness-sensitivity analysis is applied. HVAC, lighting, other electrical-load streams, and their aggregate are excluded from predictor features. The retained HVAC and lighting series are used only after forecasting for the offline opportunity calculation.

Each rolling forecast uses 96 historical 15-min intervals and predicts the next 96 intervals. Chronological train,

validation, and test segments are separated by 24.25-h gaps. This prevents a saved forecast history or target window from straddling a split boundary. It supports a clean distinction between fitting, policy selection, and held-out saved-output evaluation; it does not resolve source-level timestamp or imputation provenance.

B. Forecast Models and Validation-Selected Score Blend

The comparison includes a train-only historical weekday/time-slot average, LightGBM, random forest, a direct linear occupancy baseline (legacy artifact key DLinear), and a compact Transformer encoder (legacy artifact key Original Transformer). Table II gives the stored implementation settings. The saved outputs are bounded $[0, 1]$ Empty-class scores. They are denoted by p below for compactness, but no post-hoc calibrator is fitted and they must not be interpreted as calibrated conditional probabilities. Historical runs bear seed labels 42, 43, and 44 and their stored scores are pointwise averaged. However, deep modules were instantiated before the seed-reset function was called, so those labels do not establish controlled initialization; a corrected full retraining is required for a seed-dispersion claim.

The saved primary hybrid blends three Empty-class scores, not labels, features, or logits. For time t it is

$$p_H(t) = 0.15p_{\text{hist}}(t) + 0.60p_{\text{LGBM}}(t) + 0.25p_{\text{Tr}}(t), \quad (1)$$

where the nonnegative weights sum to one. These values are the nominal maximum of validation AUPRC on a declared 0.05 simplex grid of 231 legal weights. Test scores are loaded only after this canonical selection. No hybrid component is retrained on validation or test data. The small separation among validation candidates is evaluated explicitly in Section V-B rather than being treated as proof of a uniquely optimal blend.

C. Stable-Window Policy and Offline Opportunity

For a model-specific score threshold τ , define an initial recommendation $b_{d,t} = \mathbb{I}[p_{d,t} \geq \tau]$. Let $L = 4$ be the number of 15-min intervals in one hour. The policy recommendation $r_{d,t}$ equals one only when $b_{d,t}$ belongs to a contiguous run of at least L intervals within a midnight-labelled daily horizon. Runs are extracted independently within each 24-h forecast horizon and never join different horizons. The interval camera-label-conflict rate is

$$C = \frac{\sum_{d,t} r_{d,t}(1 - y_{d,t})}{\sum_{d,t} r_{d,t}}, \quad (2)$$

with $C = 0$ if there are no recommendations.

Let $\tilde{h}_{d,t}$ and $\tilde{\ell}_{d,t}$ denote the processed south-HVAC and south-lighting meter streams, respectively, in nominal kW. Define

$$q_{d,t} = 0.25 \max(\tilde{h}_{d,t} + \tilde{\ell}_{d,t}, 0), \quad (3)$$

which is a processed meter-derived 15-min load proxy in kWh. The nominal full-metered accounting convention uses a coefficient of one for each stream, and multiplication

TABLE II

Stored model specifications and information availability. Saved non-deterministic outputs average historical runs labelled 42, 43, and 44; deep-module construction preceded seed reset, so this is not a controlled seed-dispersion study.

Model	Inputs and training set	Stored implementation setting
Historical average	Known target weekday and 15-min slot; train partition only	Deterministic mean occupied label by weekday/time slot, with the train mean as fallback
LightGBM and random forest	Occupancy/count lags and rolling summaries; completed-input-bin sensor values; known-future calendar features; at most 200,000 stratified training rows per seed	LightGBM: 320 trees, learning rate 0.05, 31 leaves, minimum child 50, column fraction 0.85, no row bagging (subsample=1, subsample_freq=0). Random forest: 70 trees, depth 14, minimum leaf 20, square-root feature sampling
Direct linear occupancy baseline (legacy DLinear)	Past 96 occupied indicators	One direct 96-to-96 linear mapping; no trend/seasonal decomposition; maximum 12 epochs, validation-loss early stopping with patience 4
Compact Transformer encoder (legacy Original Transformer)	Past 96 occupancy/count, completed-input-bin sensor, and calendar values; known-future calendar values	One encoder layer, 32-dimensional state, 4 heads, 0.1 dropout; known-future calendar projection; AdamW at 10^{-3} ; maximum 12 epochs and validation-loss early stopping with patience 4

by 0.25 h converts kW to interval kWh. Each stream is 8.03% missing before the post-import row-order forward fill, so q includes carried-forward values and is neither a fully observed meter series nor a verified controllable load. The paper reports

$$O_{\text{safe}} = \sum_{d,t} r_{d,t} y_{d,t} q_{d,t}. \quad (4)$$

Equation (4) is an offline overlap measure: it sums the processed load proxy only where a recommendation and a subsequently observed camera-derived Empty label coincide. It neither changes a load nor estimates the counterfactual load under a control action. For each model, τ is selected from $\{0.050, 0.075, \dots, 0.950\}$ on validation midnight-labelled horizons to maximize O_{safe} subject to the empirical condition $C_{\text{val}} \leq 0.10$, breaking exact opportunity ties by Empty recall. Because O_{safe} requires realized labels and processed loads, it is an offline validation-selection objective, not an online decision-time quantity. The selected threshold is then fixed for all test policy calculations.

IV. Experimental Design

A. Evaluation Scopes and Metrics

Forecast metrics use all overlapping rolling test forecasts. The primary metric is Empty-class area under the precision-recall curve (AUPRC), which is appropriate to report alongside other measures for an imbalanced binary class [23]. Empty precision, recall, and F1 in the model-comparison table use a separate $p_{\text{Empty}} \geq 0.5$ descriptive score cutoff. They do not use the operating-policy threshold in Eq. (2). This distinction prevents a descriptive classifier cutoff from being mistaken for a selected operating rule.

Policy metrics use a different, non-overlapping scope. They use 39 validation and 43 test midnight-labelled

96-step horizons. The latter have an effective 00:15 issue boundary under the completed-bin convention in Section III. For every model, the reported threshold is selected only from validation policy horizons. The fixed test policy reports recommended, camera-label-safe, and conflict intervals, stable recommended and camera-label-safe windows, coverage, and Eq. (4). The temporal split and fixed-policy evaluation follow the general need to preserve temporal order when estimating time-series performance [24]; they do not turn the cleaned source release into an independently verified real-time data stream.

B. Uncertainty, Diagnostics, and Reproducibility

Uncertainty is quantified with 2,000 paired daily-block bootstrap resamples of the 43 test policy horizons while holding model weights and thresholds fixed. The resulting intervals are conditional on the selected model and policy. They do not account for uncertainty introduced by model selection. Brier score, log loss, and 10-bin quantile expected calibration error are reported as held-out diagnostics of the uncalibrated scores; no post-hoc calibrator is fitted. A sensitivity analysis varies the minimum window length from 15 min to 4 h while preserving the validation-selected threshold.

The saved-output path is reproducible without raw data: it validates aligned prediction exports, repeats the validation-only blend and threshold selection, and regenerates canonical result artifacts. It is not an end-to-end empirical retraining path. The raw acquisition streams, source-side imputation lineage, and source timestamp convention are absent from the repository, and the deep-model initialization ordering must be corrected before a fully controlled retraining claim. Repository tests check split separation, timestamp offsets, convex-weight selection, policy accounting, and the requirement that the two exploratory selection routines freeze candidates before their test diagnostics are read.

C. Primary and Exploratory Analyses

The canonical analysis is the validation-selected primary hybrid in Eq. (1) and the same protocol applied to fixed reference models. The repository also contains a test-ranked balanced hybrid, an expanded joint weight-threshold search, and a window-aware search. None replaces the canonical primary result. The test-ranked model is excluded from the main analysis, while the joint and window-aware candidates are reported only as status-controlled exploratory material in Appendix A. This separation is necessary because the expanded candidates use an enlarged validation search surface and their historical test diagnostics are not fresh, untouched confirmation.

V. Results

A. Forecast Quality on Overlapping Test Forecasts

Table III reports the three canonical models central to the paper. The primary hybrid has the highest all-overlap Empty AUPRC point estimate (0.8514), followed closely by the historical average (0.8497). LightGBM is 0.8382. The hybrid also has the lowest reported Brier score among these three models (0.0928), but this is a diagnostic for uncalibrated scores rather than evidence of calibrated probabilities. These values summarize 388,032 overlapping forecast rows from 4,042 anchors and should not be interpreted as 43-horizon policy estimates or as prospective real-time performance. The extended seven-model comparison is retained in Appendix A for completeness, but the test-ranked balanced hybrid is intentionally omitted from both tables.

The strong schedule baseline is a substantive result. It indicates that weekday and time-of-day structure in this building carries much of the Empty-class predictive signal. The hybrid’s point AUPRC margins are small: 0.0017 over the historical baseline and 0.0132 over LightGBM. Section V-C therefore evaluates these differences at the daily-horizon level rather than treating their ranking in Table III as decisive evidence of superiority.

B. Validation-Selection Stability

The canonical weight vector is a nominal grid maximum, not a uniquely identified optimum. Its validation AUPRC is 0.728638; the runner-up differs by only 0.000032, and eight of the 231 legal vectors lie within 0.0005. Table IV gives deliberately separate, validation-only perturbation diagnostics. The blend diagnostic restricts itself to 39 non-overlapping validation policy horizons rather than silently reusing test data, so it is not an alternative selection rule for the canonical all-overlap analysis. It nonetheless shows that ranking nearby blends is sensitive to validation-horizon perturbation. Threshold selection is also sensitive in daily-block bootstrap resamples. These diagnostics support bounded interpretation of the reported weights and threshold; they do not justify substituting a bootstrap winner after seeing the fixed test outcomes.

C. Validation-Selected Policy Outcomes

Table V and Fig. 1 report the policy scope: 43 non-overlapping test horizons and 4,128 15-min target intervals. The horizons are anchored by midnight-labelled completed input bins and have an effective 00:15 availability boundary. Every listed threshold met the empirical validation 10% interval-conflict condition. The primary hybrid used $\tau = 0.875$ after a validation conflict rate of 8.75%. On test, it recommended 259 intervals. All 259 were subsequently camera-label-empty. Those intervals formed 14 recommended windows, all camera-label-safe under the observed label, and coincided with 490.1 kWh of offline camera-label-empty load-proxy overlap (11.40 kWh/horizon-day).

LightGBM produced a nearly equal 493.9 kWh opportunity, but 11 of 265 recommended intervals were camera-labeled occupied (4.15%) and three of 19 recommended windows contained an occupancy conflict. The historical average was equally conflict-free under this label in the test period but coincided with much less processed load-proxy opportunity (94.6 kWh). Figure 1 is deliberately restricted to these three pre-specified models; it does not display test-ranked or retrospective exploratory candidates.

D. Conditional Uncertainty and Diagnostics

The daily-block uncertainty results in Table VI condition on the previously selected weights and thresholds. The primary hybrid’s daily-horizon AUPRC is 0.8522 with a 95% percentile interval of [0.7918, 0.9016]. This differs from the all-overlap value in Table III because the aggregation unit differs. The paired AUPRC intervals for primary minus LightGBM and primary minus historical average both cross zero. The paired safe-opportunity interval also crosses zero for primary minus LightGBM, although the primary-minus-historical comparison is positive over its reported interval. Figure 2 makes the contrast explicit rather than visually magnifying the small AUPRC point differences.

The primary hybrid’s held-out score diagnostics are Brier score 0.0928, log loss 0.2984, and 10-bin quantile ECE 0.0253. No recalibration was applied. At the fixed primary threshold, a one-hour or two-hour minimum window retains 490.1 kWh and zero observed camera-label conflicts, while a four-hour requirement retains 407.8 kWh and 11 fully camera-label-safe windows. These are sensitivity results at a fixed threshold, not re-optimized policies. Finally, every observed primary-hybrid camera-label conflict block is zero, so a nonparametric bootstrap also returns a zero conflict interval. Under an unverified independent-trial assumption, zero events yield one-sided 95% upper bounds of 1.15% for 259 intervals and 19.3% for 14 windows. Intervals and windows are clustered, so neither bound is a deployment-safety guarantee.

TABLE III

Canonical three-model forecast comparison on all overlapping held-out test forecasts (Empty = 1). Scores are not post-hoc calibrated. Precision, recall, and F1 use a descriptive 0.5 score cutoff. The policy thresholds in Table V are selected separately on validation.

Model	Empty AUPRC	Empty Brier	Empty precision	Empty recall	Empty F1
Historical average	0.8497	0.0974	0.7264	0.8015	0.7621
LightGBM	0.8382	0.0956	0.7610	0.7593	0.7601
Primary hybrid	0.8514	0.0928	0.7677	0.7556	0.7616

TABLE IV

Validation-only selection-stability diagnostics. The blend perturbation uses 39 non-overlapping validation policy horizons and the top 12 full-overlap grid candidates; it is descriptive and does not replace the canonical all-overlap selection. Threshold diagnostics apply the declared validation policy rule. No test export is read.

Diagnostic	Result	Frequency / value
Full-overlap 231-weight grid	Nominal rank-1 AUPRC; rank-1 minus rank-2; candidates within 0.0005	0.728638; 0.000032; 8
Daily-block bootstrap, top-12 blends (1,000)	Canonical (0.15, 0.60, 0.25) selected; most frequent (0.05, 0.80, 0.15)	23 (2.3%); 365 (36.5%)
Leave-one-validation-horizon-out, top-12 blends (39)	Canonical selected; most frequent (0.05, 0.75, 0.20)	1 (2.6%); 31 (79.5%)
Daily-block bootstrap, primary threshold (1,000)	Canonical $\tau = 0.875$; $\tau = 0.850$; $\tau = 0.950$	111 (11.1%); 376 (37.6%); 375 (37.5%)
Leave-one-validation-horizon-out, primary threshold (39)	Canonical $\tau = 0.875$	36 (92.3%)

TABLE V

Validation-selected one-hour stable-window policies evaluated on 43 non-overlapping test horizons. Anchors are midnight-labelled completed input bins with effective 00:15 boundaries. Every threshold maximizes validation safe opportunity subject to empirical validation interval conflict $\leq 10\%$. “Safe” means recommended and subsequently camera-label-empty. Opportunity is an offline processed-load-proxy overlap, not measured energy savings.

Model	τ	Val. conflict	Test conflict	Safe opportunity	Safe/rec. intervals	Safe/rec. windows	kWh/day
Historical average	0.950	8.33%	0.00% (0/66)	94.6	66/66	12/12	2.20
LightGBM	0.950	9.61%	4.15% (11/265)	493.9	254/265	16/19	11.49
Primary hybrid	0.875	8.75%	0.00% (0/259)	490.1	259/259	14/14	11.40

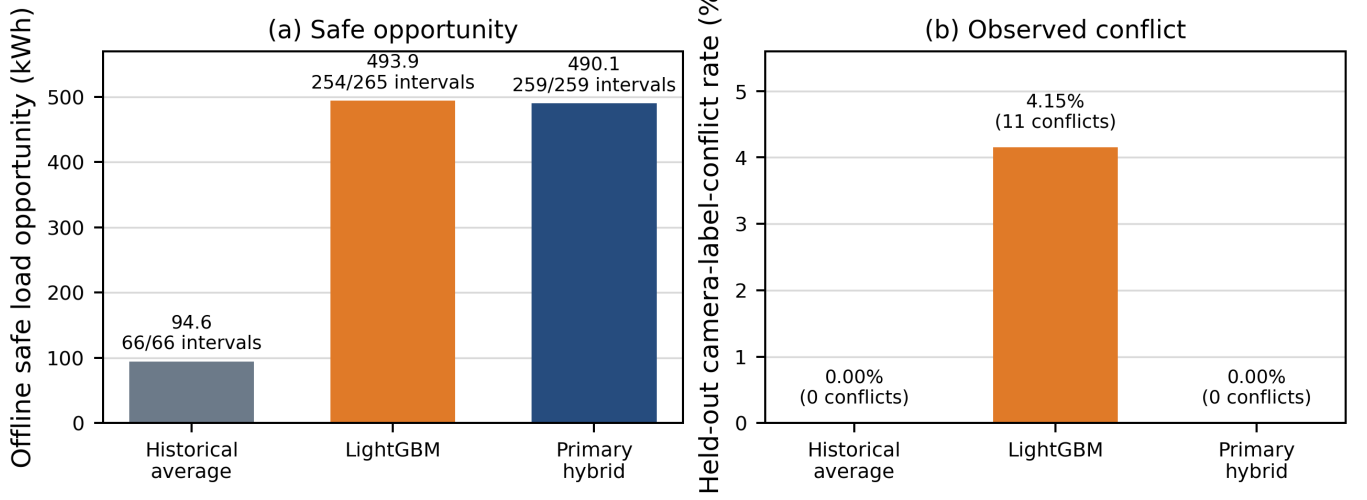


Fig. 1. Fixed held-out policy outcomes for the schedule baseline, LightGBM, and the validation-selected primary hybrid. Thresholds were selected on 39 validation daily horizons to maximize offline safe load opportunity subject to an empirical interval conflict rate no greater than 10%; the plotted values are one post-bin evaluation on 43 test horizons. “Safe” means recommended and subsequently camera-label-empty, not physical absence, a verified load shift, or energy saving.

VI. Discussion

A. What the Comparison Does and Does Not Show

The historical average’s performance is informative rather than an inconvenience to be optimized away. A

TABLE VI

Conditional daily-block uncertainty from 2,000 paired bootstrap resamples of 43 non-overlapping test horizons. Daily-horizon AUPRC is not the all-overlap AUPRC in Table III. All intervals condition on selected weights and thresholds.

Estimate type	Quantity	Point estimate	95% low	95% high
Primary model	Daily-horizon Empty AUPRC	0.8522	0.7918	0.9016
Primary model	Safe opportunity (kWh)	490.1	153.1	966.7
Paired contrast	Primary – LightGBM AUPRC	+0.0163	-0.0050	+0.0426
Paired contrast	Primary – historical AUPRC	+0.0042	-0.0304	+0.0377
Paired contrast	Primary – LightGBM opportunity (kWh)	-3.8	-208.2	+174.6
Paired contrast	Primary – historical opportunity (kWh)	+395.5	+127.0	+786.2

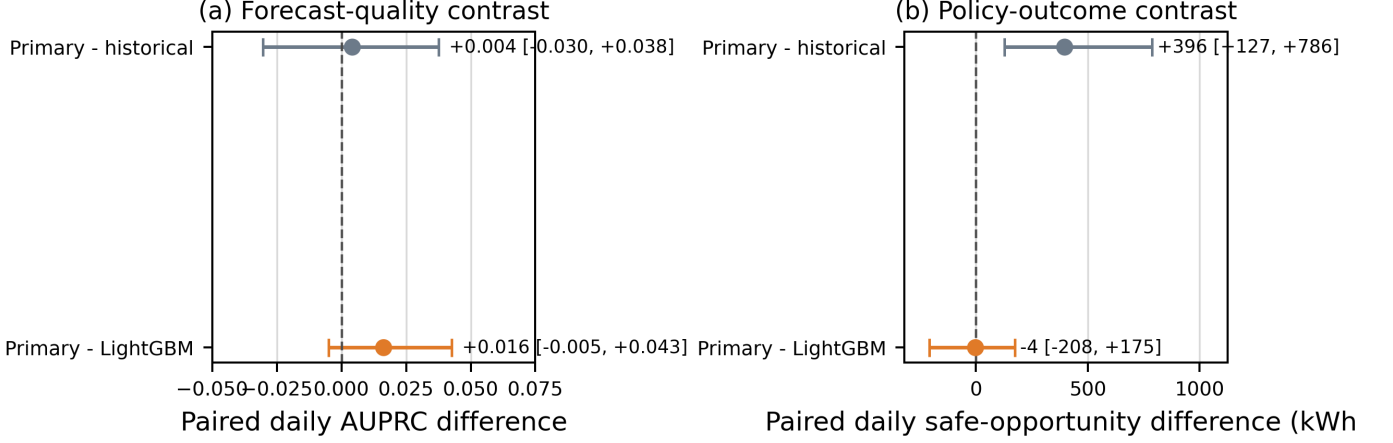


Fig. 2. Paired 95% daily-block bootstrap intervals (2,000 resamples) for the primary hybrid minus each canonical baseline. The dashed line denotes no difference. Intervals condition on fixed, validation-selected model weights and operating thresholds; they do not include model-selection uncertainty.

stable schedule can produce a highly competitive Empty forecast in a single office building, and the small point AUPRC advantage of the primary hybrid is not decisive under the reported paired daily-block intervals. The hybrid’s practical distinction in this saved test period is not a large forecast-score gain: it selects an operating point with approximately the same processed load-proxy opportunity as LightGBM and no camera-labeled occupied recommendations. This is a reason to report forecast and policy metrics together, not a reason to declare a universally superior model.

The offline opportunity formulation also provides a useful guardrail for interpretation. It makes the recommendation rule, duration requirement, conflict denominator, and load proxy inspectable. In particular, it avoids calling every correctly predicted Empty interval operationally useful. A short isolated interval is removed by the stable-window rule. At the same time, Eq. (4) has no counterfactual. The processed HVAC and lighting values may not be controllable at that moment, shifting a load may create a later penalty, and a real intervention would require constraints on equipment, indoor conditions, tariffs, and occupants. This paper therefore does not estimate savings, flexibility capacity, carbon reduction, or comfort preservation.

B. Validity Limits

The temporal and provenance limitation is fundamental. A stored 00:00 anchor contains a completed [00:00, 00:15) input bin, so the current evidence supports only an effective 00:15 post-bin boundary and targets [00:15, next 00:15). It does not establish a midnight-issued or real-time recommendation. Moreover, the repository starts from the cleaned release; its documented upstream cleaning can include non-local imputation, and neither original raw streams nor observation-end timestamps are committed. Row-order forward fill performed after import cannot repair that provenance gap. Thus, the saved output is internally ordered but cannot be called fully prospective or source-causally leakage-free.

Several additional limitations constrain the result. The data cover selected south zones in one building and a 43-horizon test policy period; external and cross-building validation are unavailable. The camera-count label can miss occupants and does not establish physical absence. Results called camera-label-safe or conflict-free apply only to that label. Wi-Fi has substantial pre-fill missingness without an age cap, and the HVAC and lighting streams used in q are each 8.03% missing before row-order filling. The study does not establish robustness to alternative

sensor availability or load-imputation patterns. The source timezone is inferred through a solar-pattern diagnostic rather than explicitly documented.

The statistical limitations are also material. The bootstrap conditions on selected weights and thresholds and cannot generate unseen conflicts when all resampled blocks have zero conflicts. The nominal blend winner exceeds the runner-up by only 0.000032 validation AUPRC, and the validation-only perturbation diagnostics in Table IV demonstrate selection instability. Deep-model seed labels do not control initialization because construction preceded seed reset. Finally, saved exports do not contain aligned per-seed predictions or all raw source files, so seed dispersion and end-to-end retraining cannot be reconstructed. The available artifacts verify a post-bin validation-before-test selection procedure for the canonical saved-output analysis, but they are not a preregistered, fully blinded confirmatory evaluation with a fresh test firewall.

C. Status of Expanded Searches and Next Evidence

The repository’s decision-aware joint search and window-aware search are not evidence for replacing the canonical policy. Their code now freezes validation candidates before reading the test diagnostic, but both enlarge the search surface and their historical test diagnostics are not fresh untouched confirmation. Appendix A labels their point estimates as exploratory or as frozen future challengers rather than performance improvements. This distinction is essential because decision-focused learning would require downstream-objective training, which this pipeline does not perform [19], [20].

The next defensible empirical experiment is more specific than simply rerunning the saved script: obtain source streams with observation-end timestamps and imputation lineage; declare a bin-end/issue-time convention; correct seed initialization before model construction; lock package versions; retrain without consulting future test outcomes; and apply the frozen policy to a later untouched period or a new building. A simulator or intervention with explicit comfort and equipment constraints is then needed to test actual control and energy claims. That sequence would make it possible to evaluate the operational questions addressed by occupancy-aware HVAC and ventilation studies [7], [17], [18].

VII. Conclusion

This paper presents a transparent offline, post-bin assessment from saved next-24-h occupancy scores to stable Empty-window recommendations. In one Building 59 case study, a validation-selected historical–LightGBM–compact-Transformer score blend achieved 0.8514 all-overlap Empty AUPRC across 388,032 test rows and, at its fixed one-hour policy, 490.1 kWh of offline safe load opportunity with zero observed camera-label conflicts across 259 recommended test intervals. The historical schedule baseline remained nearly equal in AUPRC, and

the primary hybrid’s paired differences from LightGBM and the schedule baseline were not decisive for forecast quality or for opportunity versus LightGBM. Validation perturbations further show that the selected blend and threshold should not be treated as uniquely optimal.

The contribution is therefore a reproducible saved-output forecast-to-policy assessment with explicit evidence boundaries. It does not demonstrate energy savings, comfort preservation, physical absence, controller safety, calibrated risk, real-time availability, or generalization. The left-labelled bin timing and cleaned-source provenance mean that an operational claim requires an empirical rerun with raw/provenance-tagged streams, corrected deep-model seed initialization, a locked protocol, and a later untouched evaluation period before submission as a prospective operational method.

Appendix A

Extended Results and Exploratory-Analysis Status

Table VII preserves the complete canonical comparison set. The test-ranked balanced hybrid is not shown because its architecture was selected after inspection of test ranking and is supplementary by definition. The primary hybrid remains the nominal validation-selected canonical model in the manuscript.

The repository additionally contains a joint weight-threshold search over 231 convex blend weights and 37 score thresholds (8,547 validation pairs) and a window-aware search over the resulting validation surface. After the present code audit, both routines freeze validation candidates before loading their test diagnostics. They nevertheless enlarge the search surface, use historical results whose test period is no longer fresh confirmation, and inherit the post-bin/source-provenance limits of the base exports. The decision-optimal current-test point estimates (623.5 and 650.4 kWh for two joint-search variants) and the window-aware challenger’s current-test point estimate (511.6 kWh with 2.49% interval conflict) are therefore retrospective diagnostics, not headline efficacy results.

a) Status controls.: The test-ranked balanced hybrid is supplementary only because its architecture family was selected after inspection of test ranking. The joint search is exploratory and cannot replace the canonical primary policy. The window-aware candidate is a frozen future challenger, not a confirmed result. None of these searches trains a downstream decision loss. A future evaluation may use a frozen candidate and its declared constraints on a later untouched chronological period, ideally with control, comfort, and counterfactual energy measurements.

Data and Code Availability

The cleaned Building 59 data package is available from Dryad [22]; original acquisition streams, source-side imputation lineage, and timestamp provenance are not committed to this repository. The submitted artifact should include the repository revision, the canonical saved

TABLE VII

Extended canonical model comparison on all overlapping held-out test forecasts (Empty = 1). Values are uncalibrated-score diagnostics; the test-ranked balanced hybrid is excluded because it is supplementary only.

Model	Empty AUPRC	Empty precision	Empty recall	Empty F1
Historical average	0.8497	0.7264	0.8015	0.7621
LightGBM	0.8382	0.7610	0.7593	0.7601
Random forest	0.8329	0.7510	0.8019	0.7756
Compact Transformer encoder (legacy Original Transformer)	0.7621	0.7253	0.6022	0.6581
Direct linear occupancy baseline (legacy DLinear)	0.5933	0.5764	0.6657	0.6179
Seasonal-compact-encoder blend	0.8490	0.7465	0.7408	0.7436
Primary hybrid	0.8514	0.7677	0.7556	0.7616

prediction and result exports, and the scripts used to regenerate the manuscript figures and tables. A persistent archival URL must be inserted here before submission.

Author Contributions, Funding, and Competing Interests

To preserve anonymous review, contributor identities, funding information, and conflict-of-interest declarations are omitted from this draft. The corresponding author must replace this statement with the target venue's required declarations before submission.

References

- [1] Y. Chen, P. Xu, J. Gu, F. Schmidt, and W. Li, "Measures to improve energy demand flexibility in buildings for demand response (DR): A review," *Energy and Buildings*, vol. 177, pp. 125–139, 2018.
- [2] R. G. Junker, A. G. Azar, R. A. Lopes, K. B. Lindberg, G. Reynnders, R. Relan, and H. Madsen, "Characterizing the energy flexibility of buildings and districts," *Applied Energy*, vol. 225, pp. 175–182, 2018.
- [3] V. L. Erickson, M. A. Carreira-Perpiñán, and A. E. Cerpa, "Occupancy modeling and prediction for building energy management," *ACM Transactions on Sensor Networks*, vol. 10, no. 3, 2014, art. no. 42, 28 pp.
- [4] Y. Jin, D. Yan, A. Chong, B. Dong, and J. An, "Building occupancy forecasting: A systematical and critical review," *Energy and Buildings*, vol. 251, p. 111345, 2021.
- [5] J. Caballero-Peña, G. Osma-Pinto, J. M. Rey, S. Nagarsheth, N. Henao, and K. Agbossou, "Analysis of the building occupancy estimation and prediction process: A systematic review," *Energy and Buildings*, vol. 313, p. 114230, 2024.
- [6] T. Li, X. Liu, G. Li, X. Wang, J. Ma, C. Xu, and Q. Mao, "A systematic review and comprehensive analysis of building occupancy prediction," *Renewable and Sustainable Energy Reviews*, vol. 193, p. 114284, 2024.
- [7] Y. Peng, A. Rysanek, Z. Nagy, and A. Schlüter, "Using machine learning techniques for occupancy-prediction-based cooling control in office buildings," *Applied Energy*, vol. 211, pp. 1343–1358, 2018.
- [8] M. Esrafilian-Najafabadi and F. Haghighat, "Impact of occupancy prediction models on building HVAC control system performance: Application of machine learning techniques," *Energy and Buildings*, vol. 257, p. 111808, 2022.
- [9] G. Lin, A. Casillas, M. Sheng, and J. Granderson, "Performance evaluation of an occupancy-based HVAC control system in an office building," *Energies*, vol. 16, no. 20, p. 7088, 2023.
- [10] Z. Li and B. Dong, "Short term predictions of occupancy in commercial buildings—performance analysis for stochastic models and machine learning approaches," *Energy and Buildings*, vol. 158, pp. 268–281, 2018.
- [11] S. H. Ryu and H. J. Moon, "Development of an occupancy prediction model using indoor environmental data based on machine learning techniques," *Building and Environment*, vol. 107, pp. 1–9, 2016.
- [12] K. Sun, I. Qaisar, M. A. Khan, T. Xing, and Q. Zhao, "Building occupancy number prediction: A transformer approach," *Building and Environment*, vol. 244, p. 110807, 2023.
- [13] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "LightGBM: A highly efficient gradient boosting decision tree," in *Advances in Neural Information Processing Systems*, vol. 30, 2017, pp. 3146–3154.
- [14] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [15] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems*, vol. 30, 2017, pp. 5998–6008.
- [16] A. Zeng, M. Chen, L. Zhang, and Q. Xu, "Are transformers effective for time series forecasting?" *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 37, no. 9, pp. 11 121–11 128, 2023.
- [17] C. Zhuang, R. Choudhary, and A. Mavrogianni, "Probabilistic occupancy forecasting for risk-aware optimal ventilation through autoencoder bayesian deep neural networks," *Building and Environment*, vol. 219, p. 109207, 2022.
- [18] T. Wietzke, J. Gall, and K. Graichen, "Occupancy prediction for building energy systems with latent force models," *Energy and Buildings*, vol. 307, p. 113968, 2024.
- [19] B. Wilder, B. Dilkina, and M. Tambe, "Melding the data-decisions pipeline: Decision-focused learning for combinatorial optimization," *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 33, no. 1, pp. 1658–1665, 2019.
- [20] A. N. Elmachtoub and P. Grigas, "Smart "predict, then optimize"," *Management Science*, vol. 68, no. 1, pp. 9–26, 2022.
- [21] N. Luo, Z. Wang, D. Blum, C. Weyandt, N. Bourassa, M. A. Piette, and T. Hong, "A three-year dataset supporting research on building energy management and occupancy analytics," *Scientific Data*, vol. 9, no. 1, p. 156, 2022.
- [22] T. Hong, N. Luo, D. Blum, and Z. Wang, "A three-year building operational performance dataset for informing energy efficiency," *Dryad Dataset*, ver. 6, 2022. [Online]. Available: <https://doi.org/10.7941/D1N33Q>
- [23] T. Saito and M. Rehmsmeier, "The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets," *PLOS ONE*, vol. 10, no. 3, p. e0118432, 2015.
- [24] V. Cerqueira, L. Torgo, and I. Mozetič, "Evaluating time series forecasting models: An empirical study on performance estimation methods," *Machine Learning*, vol. 109, no. 11, pp. 1997–2028, 2020.